Probability & Statistics (MT-2005)

Descriptive Statistics

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Tentative Marks Distribution & Text Books

Assessment Type	Weightages
Quiz	10%
Homeworks	15%
Sessional I	15%
Sessional II	15%
Final Exam	45%

* Absolute grading policy would be followed

Text Books

- Elementary Statistics, A step by Step Approach, Allan G. Bluman
- Probability and Statistics for Engineers and Scientists by Walpole, Myers, and Myers.

Course Plan Contn'd...

Reference Books

- Introduction to Probability by Dimitri P. Bertsekas and John N. Tsitsiklis.
- Think Stats: Exploratory Data Analysis in Python by Allen B. Downey.
- The Elements of Statistical Learning by Hastie, Tibshirani, and Friedman

What is Probability and Statistics

Probability

Probability is a measure of the likelihood that an event will occur, expressed as a value between 0 and 1, where:

- 0 means the event is impossible.
- 1 means the event is certain.

It is mathematically given by:

$$P = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Statistics

Statistics is the branch of mathematics that involves collecting, organizing, analyzing, interpreting, and presenting data to draw meaningful conclusions and make informed decisions.

What is Probability and Statistics Contn'd...

Difference Between Probability and Statistics

- **Probability**: Starts with a model to predict the likelihood of future outcomes (predictive, theory-to-data).
- **Statistics**: Starts with data to infer parameters or understand the underlying model (inferential, data-to-theory).

Similarities Between Probability and Statistics

Both deal with uncertainty and use mathematical frameworks to analyze and make decisions about random phenomena. They rely on concepts like events, distributions, and outcomes.

Branches of Statistics

The study of statistics has two major branches depending on how data are used: **descriptive statistics** and **inferential statistics**.

1. Descriptive Statistics: Focuses on summarizing and describing the main features of a data set. Methods include:
   - **Data visualization:** Tables, Graphs, charts.
   - **Numerical descriptive measures:** Measures of central tendency, Measures of variability, Measures of position.
2. Inferential Statistics Concerned with making predictions or inferences about a population based on a sample. It involves techniques such as:
   - Hypothesis testing.
   - Confidence intervals.
   - Regression analysis.

   Inferential statistics uses probability theory to estimate population parameters.

Essential Statistical Terminologies

- **Population:** The entire group of individuals, items, or events that you want to study or make inferences about.
- **Sample:** A subset of the population selected for analysis, intended to represent the population.
- **Variable:** A symbol or placeholder that can take any value from a set of possible values.
- **Random Variable:** A function that assigns a numerical value to each outcome of a random experiment.
- **Prameter:** A numerical value that describes a characteristic of a population.
- **Statistic:** A numerical value that describes a characteristic of a sample.

Essential Statistical Terminologies Contn'd...

- **Deterministic Model**: Always produces the same outcome for a given input. No randomness or uncertainty involved.
  - Example: $s = ut + \frac{1}{2}at^2$ (predicts exact position of an object under uniform acceleration).
- **Probabilistic Model**: Involves uncertainty and describes outcomes using probabilities.
  - Example: Predicting the weather, with $P(\text{Rain}) = 0.7$ and $P(\text{No Rain}) = 0.3$.
- **Accuracy:** Accuracy refers to how close a measured value is to the true value. Accuracy is concerned with the systematic errors or biases in measurements or predictions.
- **Precision:** Precision, on the other hand, refers to the consistency or repeatability of the measurements, or how close the measured values are to each other. Precision focuses on the random errors and how spread out the measurements are.

- **Outlier:** A data point or observation that differs significantly from other observations in a dataset. It is an unusually large or small value that does not fit the general pattern or trend of the data.

Data and Data Types

- **By Nature:**
  - **Qualitative (Categorical)::**
    - **Nominal:** Categories without order (e.g., Gender: Male, Female).
    - **Ordinal:** Ordered categories without equal intervals (e.g., Education Level: High School ¡ Bachelor ¡ Master).
  - **Quantitative Numerical)::**
    - **Discrete:** Countable values (e.g., Number of students).
    - **Continuous:** Measurable values (e.g., Height, Temperature).

Levels of Measurement

- **Interval:**
  - Ordered, equal intervals, no true zero.
  - Examples: Temperature, IQ scores.
- **Ratio:**
  - Ordered, equal intervals, true zero exists.
  - Examples: Height, Time.

More on Data Types

- **Univariate Data:**
  - Single variable focus (e.g., Heights of students).
  - Analysis: Mean, Median, SD.
- **Bivariate Data:**
  - Two variables, relationship analysis (e.g., Height vs Weight).
  - Techniques: Correlation, Regression.
- **Multivariate Data:**
  - More than two variables (e.g., Age, Height, Income).
  - Techniques: PCA, MANOVA.

More on Data Types

- **Rectangular Data:**
  - Organized in tables (e.g., Students' grades).
- **Non-Rectangular Data:**
  - Unstructured/Semi-structured (e.g., Emails, Hierarchical data).
- **Cross-Sectional Data:**
  - Snapshot data at a specific point in time.
- **Time-Series Data:**
  - Data over regular intervals (e.g., Monthly sales).
- **Panel Data:**
  - Combines cross-sectional and time-series data.
  - Tracks subjects over time (e.g., Health studies).

Sampling Techniques & Collection of Data

Sampling Techniques: Collection of Data

Data Collection Methods

- **Primary**: Collected directly from experiments or surveys/interviews.
- **Secondary**: Extracted from existing sources (e.g., government databases).

Sampling Techniques Statisticians use four basic methods of sampling:

1. Random Sampling: Each individual in the population has an equal chance of being selected.
    - **Method**: Random selection is typically achieved using a random number generator or drawing lots.
    - **Example**: Selecting 10 students randomly from a list of 100 students.
2. Systematic Sampling: Selects individuals at regular intervals from a sorted list.
    - **Method**: Choose a starting point randomly, then pick every k-th individual (e.g., every 5th person).
    - **Example**: In a group of 1,000, choosing every 10th person after randomly selecting a starting point.

3. Stratified Sampling: The population is divided into subgroups (strata) based on a characteristic, and random samples are drawn from each stratum.
    - **Method**: Proportional or equal representation is ensured from each subgroup.
    - **Example**: Surveying students by dividing them into strata based on grades.
4. Cluster Sampling: The population is divided into clusters, and a random selection of clusters is made. All individuals in the selected clusters are included in the sample.
    - **Method**: Often used when the population is geographically dispersed.
    - **Example**: Selecting schools randomly in a district and surveying all students in the chosen schools.

Representation of Data

Data representation is crucial for understanding sample data. Two key components of data representation:

1. Data Organization: Organizing data helps to visualize patterns and analyze relationships
   - Organization of Qualitative Data (The Categorical Frequency Distribution, The Contingency Table)
   - Organizing Quantitative Data (The Frequency Distribution(Grouped, Ungrouped)
2. Data Visualization: Visualizing data reveals patterns and relationships.
   - Visualization of Qualitative Data (Bar Chart, Multiple Bar Chart, Component Bar Chart, Pie Chart)
   - Visualization of Quantitative Data (Histogram, Frequency Polygon, Ogive)

Qualitative Data Organization and Visualization

Example # 1: The Categorical Frequency Distribution

The following list represents the recorded software errors observed during testing:

- **S**: Syntax Errors
- **R**: Runtime Errors
- **L**: Logical Errors

$$\{S, R, S, L, R, S, S, L, R, S, L, R, S, S, S, R, L, S, S, R, R, S, L,$$
$$S, R, L, R, S, R, S, L, L, S, S, R, S, R, L, R, S, R, L, S, L, S,$$
$$R, L, S, L, S, R, L, R, R, L, S, L, S, S\}$$

Make the table that summarizes the frequency distribution of three types of software errors along with their tally, relative frequencies percentages, where

Relative Frequency Percentages $= \frac{\text{Frequency of a Category}}{\text{Total Frequency}} \times 100$

Organization of Qualitative Data: Contn'd...

Error Type	**Tally**	**Frequency**	**Percentage**
Syntax Errors	$\cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}$	25	41.67%
Runtime Errors	$\cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}$	20	33.33%
Logical Errors	$\cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}, \cancel{\vert\vert\vert\vert}$	15	25.00%
Total	-	60	100.00%

Table: Frequency Distribution of Software Errors

Python Code: Bar Chart Example

Here is the Python code for creating a bar chart:

```python
import matplotlib.pyplot as plt

# Data for bar chart
error_types = ['Syntax Errors', 'Runtime Errors', 'Logical
    Errors']
frequencies = [45, 50, 20]

# Create bar chart
plt.bar(error_types, frequencies, color=['blue', 'green', '
    orange'])
plt.title('Frequency of Programming Errors')
plt.xlabel('Error Type')
plt.ylabel('Frequency')

# Save the plot
plt.savefig('bar_chart.png')
plt.close()
```

Bar Chart Visualization

After running the code, you will get the following bar chart:

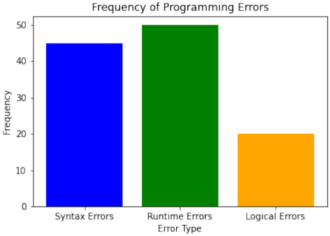


Figure: Bar Chart: Errors Across Operating Systems

Organization of Qualitative Data: Contn'd...

Example # 2:Contingency table: is a type of data table that summarizes the relationship between two categorical variables. Each cell in the table represents the frequency (or count) of observations for a specific combination of the variables.
The table below shows the distribution of different types of programming errors across various operating systems during testing:

Error Type / OS	**Windows**	**Linux**	**MacOS**	**Total**
Syntax Errors	20	15	10	45
Runtime Errors	10	25	15	50
Logical Errors	5	10	5	20
Total	35	50	30	115

Table: Contingency Table: Errors by Operating System

Python Code: Multiple Bar Chart Example

Here is the Python code for creating a multiple bar chart:

```
import matplotlib.pyplot as plt
import numpy as np

# Data for multiple bar chart
os_types = ['Windows', 'Linux', 'MacOS']
syntax_errors = [20, 15, 10]
runtime_errors = [10, 25, 15]
logical_errors = [5, 10, 5]

# Bar width and positions
bar_width = 0.2
x = np.arange(len(os_types))
```

Python Code: Multiple Bar Chart Contn'd...

```
# Plot multiple bars
plt.bar(x - bar_width, syntax_errors, width=bar_width, label
    ='Syntax Errors', color='blue')
plt.bar(x, runtime_errors, width=bar_width, label='Runtime
    Errors', color='green')
plt.bar(x + bar_width, logical_errors, width=bar_width,
    label='Logical Errors', color='orange')

# Customize chart
plt.title('Programming Errors by Operating System')
plt.xlabel('Operating System')
plt.ylabel('Frequency')
plt.xticks(x, os_types)
plt.legend()

# Save the plot
plt.savefig('multiple_bar_chart.png')
plt.close()
```

Multiple Bar Chart Visualization

After running the code, you will get the following multiple bar chart:

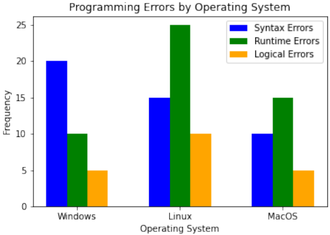


Figure: Multiple Bar Chart: Errors Across Operating Systems

Python Code: Component/Stacked Bar Chart

```python
import matplotlib.pyplot as plt
import numpy as np

# Data for stacked bar chart
os_types = ['Windows', 'Linux', 'MacOS']
syntax_errors = [20, 15, 10]
runtime_errors = [10, 25, 15]
logical_errors = [5, 10, 5]

# Plot stacked bars
plt.bar(os_types, syntax_errors, label='Syntax Errors',
    color='blue')
plt.bar(os_types, runtime_errors, bottom=syntax_errors,
    label='Runtime Errors', color='green')
plt.bar(os_types, logical_errors, bottom=np.add(
    syntax_errors, runtime_errors), label='Logical Errors',
    color='orange')
```

Python Code: Component Bar Chart Contn'd...

```
# Customize chart
plt.title('Stacked Bar Chart of Programming Errors by OS')
plt.xlabel('Operating System')
plt.ylabel('Frequency')
plt.legend()

# Save the plot
plt.savefig('stacked_bar_chart.png')
plt.close()
```

Component/Stacked Bar Chart Visualization

After running the code, you will get the following Component/Stacked Bar Chart:

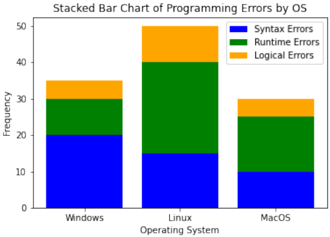


Figure: Stacked Bar Chart: Errors Across Operating Systems

Example # 3 Pie Chart: The table below shows the distribution of students across various Computer Science specializations:

Specialization	**Number of Students**
Artificial Intelligence	120
Cybersecurity	80
Data Science	100
Software Engineering	150
Human-Computer Interaction	50
Total	**400**

Table: Distribution of Students Across CS Specializations

Represent the data using Pie Chart

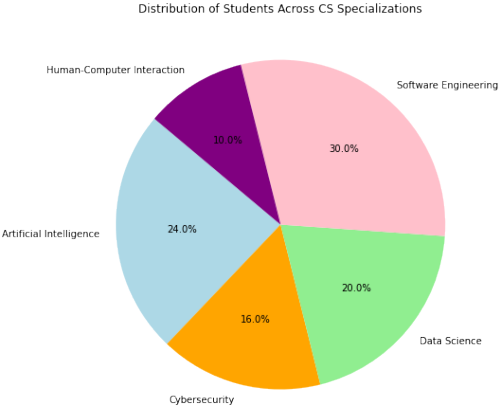
Python Code: Pie Chart

```python
import matplotlib.pyplot as plt

# Data
specializations = ['Artificial Intelligence', 'Cybersecurity', 'Data Science',
                   'Software Engineering', 'Human-Computer Interaction']
students = [120, 80, 100, 150, 50]
colors = ['lightblue', 'orange', 'lightgreen', 'pink', 'purple']  # Custom colors

# Create the pie chart
plt.figure(figsize=(8, 8))
plt.pie(students, labels=specializations, autopct='%1.1f%%',
    startangle=140, colors=colors)
```

```
# Add title
plt.title('Distribution of Students Across CS
    Specializations')


# Save the plot
plt.savefig('pie.png')
plt.close()
```

Pie Chart Visualization

After running the code, you will get the following Pie Bar Chart:



Example # 3 Pareto Chart: A Pareto chart is a combination of a bar chart and a line graph, where the bars represent frequencies (or counts), and the line represents the cumulative percentage. It's commonly used to identify the most significant factors in a dataset.
The following data represents the frequency of various types of software bugs during a project:

Bug Type	**Number of Bugs**
Syntax Errors	40
Runtime Errors	30
Logical Errors	20
UI/UX Issues	10
Performance Issues	5
Total	**400**

Represent the data using Pareto Chart

Python Code: Pareto Chart

```
import matplotlib.pyplot as plt
import numpy as np

# Data
bug_types = ['Syntax Errors', 'Runtime Errors', 'Logical
    Errors', 'UI/UX Issues', 'Performance Issues']
frequencies = [40, 30, 20, 10, 5]

# Sort data in descending order (if not already sorted)
sorted_indices = np.argsort(frequencies)[::-1]
bug_types = [bug_types[i] for i in sorted_indices]
frequencies = [frequencies[i] for i in sorted_indices]

# Calculate cumulative percentages
cumulative_percentages = np.cumsum(frequencies) / sum(
    frequencies) * 100

# Create the bar chart
fig, ax1 = plt.subplots(figsize=(10, 6))
```

Python Code: Pareto Chart Contn'd...

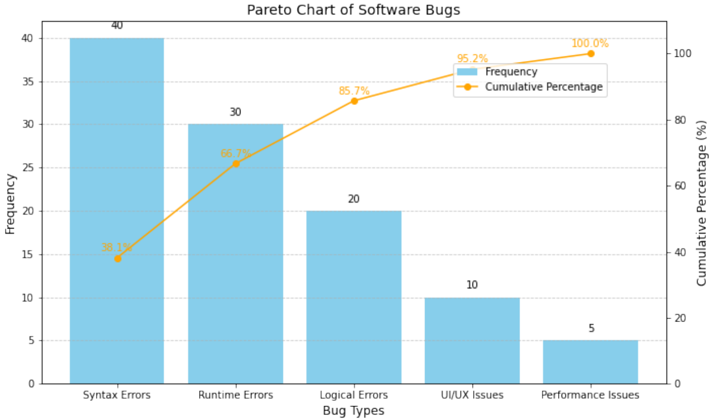
```
bars = ax1.bar(bug_types, frequencies, color='skyblue',
    label='Frequency')
ax1.set_ylabel('Frequency', fontsize=12)
ax1.set_xlabel('Bug Types', fontsize=12)
ax1.set_title('Pareto Chart of Software Bugs', fontsize=14)

# Add cumulative percentage line
ax2 = ax1.twinx()
ax2.plot(bug_types, cumulative_percentages, color='orange',
    marker='o', label='Cumulative Percentage')
ax2.set_ylabel('Cumulative Percentage (%)', fontsize=12)
ax2.set_ylim(0, 110)
```

Python Code: Pareto Chart Contn'd...

```
# Add gridlines and annotations
ax1.grid(axis='y', linestyle='--', alpha=0.7)
for i, bar in enumerate(bars):
    ax1.text(bar.get_x() + bar.get_width() / 2, bar.
        get_height() + 1, f'{frequencies[i]}', ha='center',
        fontsize=10)
for i, pct in enumerate(cumulative_percentages):
    ax2.text(i, pct + 2, f'{pct:.1f}%', ha='center',
        fontsize=10, color='orange')

# Show legend
fig.legend(loc='upper right', bbox_to_anchor=(0.85, 0.85),
    fontsize=10)
plt.tight_layout()

# Save the plot
plt.savefig('pareto.png')
plt.close()
```

Pareto Chart Visualization

After running the code, you will get the following Pareto Chart:



Quantitative Data Organization and Visualization

The scores of 30 students in a mathematics exam are given below. Represent the data distribution using a histogram.

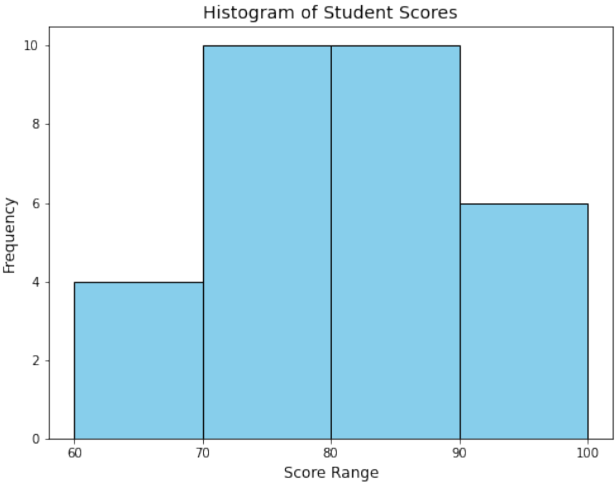
Scores of Students
75, 88, 92, 70, 81, 79, 85, 73, 95, 68,
82, 78, 80, 72, 77, 84, 69, 74, 83, 71,
90, 87, 76, 89, 93, 86, 91, 94, 67, 66

The scores of 30 students in a mathematics exam are grouped into score ranges. Below is the frequency table showing the distribution of scores:

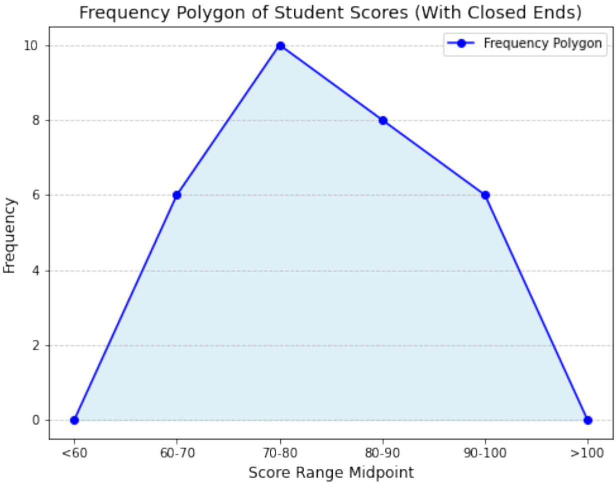
Score Range	**Frequency**	**Midpoint**	**Cumulative Frequency**
60–70	6	65	6
70–80	10	75	16
80–90	8	85	24
90–100	6	95	30

Histogram

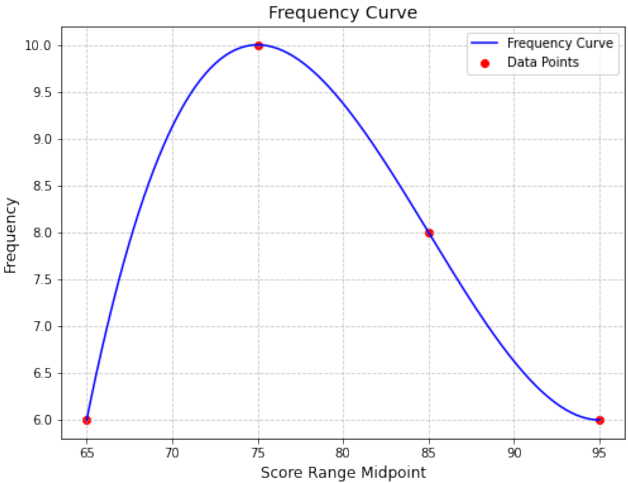
Below is the histogram showing the distribution of student scores:



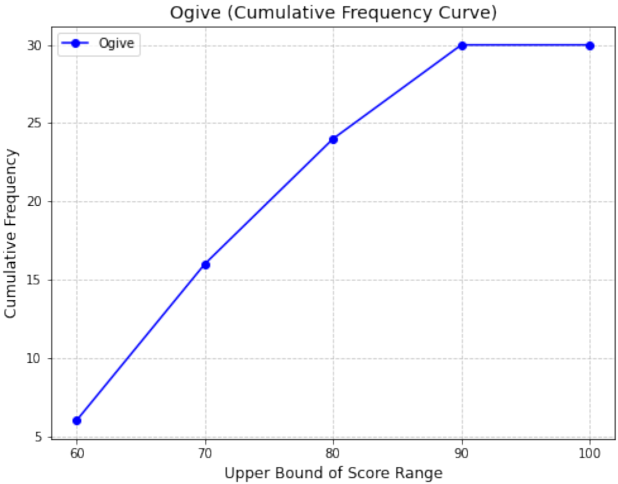
Frequency Polygon



Frequency Curve



Ogive


Ogive (Cumulative Frequency Curve)
Ogive
Cumulative Frequency
Upper Bound of Score Range
30
25
20
15
10
5
60
70
80
90
100

Single Stem-and-Leaf Plot

The following is the stem-and-leaf plot for the dataset: $\{65, 67, 69, 72, 73, 76, 77, 78, 81, 84, 85, 88, 92, 95\}$.

Stem	Leaves
6	5, 7, 9
7	2, 3, 6, 7, 8
8	1, 4, 5, 8
9	2, 5

Back-to-Back Stem-and-Leaf Plot

Dataset 1: $\{67,72,78,65,88,85,82,95,92,76,84,73,77,81,69\}$
Dataset 2: $\{66,70,74,80,83,87,89,91,93,96,75,78,82,86,90\}$
We compare two datasets using a back-to-back stem-and-leaf plot:

Leaves (Dataset 2)	Stem	Leaves (Dataset 1)
6, 6	6	5, 7, 9
0, 4, 5, 7	7	2, 3, 6, 7, 8
0, 2, 6	8	1, 2, 4, 5, 8
0, 1, 3, 6	9	2, 5

Legend:

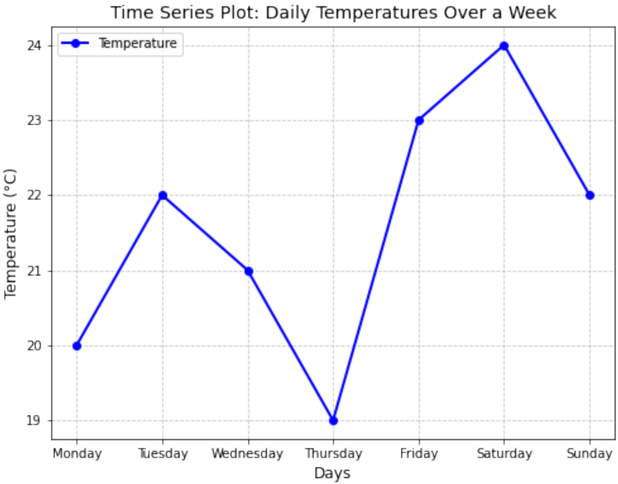
- **Stem:** Represents the tens digit.
- **Leaves:** Represents the units digit.

Daily Temperatures Over a Week

The following table shows the daily temperatures (in $^\circ$C) recorded over a week:

Day	**Temperature ($^\circ$C)**
Monday	20
Tuesday	22
Wednesday	21
Thursday	19
Friday	23
Saturday	24
Sunday	22

Time Series Graph



Numerical Descriptive Statistics

Numerical Descriptive Statistics: Key Points

- **Purpose of Numerical Measures:**
  - Convey a mental image of data, phenomena, or objects.
  - Provide a more concise description compared to graphical methods.
- **Limitations of Graphical Methods:**
  - Inappropriate for statistical inference.
  - Difficult to compare sample and population frequency histograms.
- **Data Reduction:**
  - Reduces the number of entries describing a dataset.
  - Comes with progressive loss of information.
  - Requires identifying the most important characteristics of the distribution.
- **Focus of Numerical Descriptive Statistics:**
  - Summarize main features of a dataset.
  - Provide insights into:
    - Central tendency.
    - Variability.
    - Position.
    - Shape of data.

Measures of Central Tendency: Arithmetic Mean

Measures of Central Tendency: Arithmetic Mean

Measures of central tendency provide a single value that summarizes or represents the central point of a dataset. These measures help describe the overall characteristics of data and are widely used in statistics, data analysis, and research.

- Mean
- Median
- Mode

Ungrouped Data For a dataset $x_1, x_2, \ldots, x_n$:

$$\text{Mean} = \bar{x} = \frac{\displaystyle\sum_{i=1}^{n} x_i}{n}$$

where n is the number of observations.

Grouped Data

$$\text{Mean} = \bar{x} = \frac{\sum_{i=1}^{k} f_i x_i}{\sum_{i=1}^{k} f_i}$$

where f_i is the frequency of the i-th class, x_i is the midpoint of the i-th class, and k is the number of classes.

Properties of Arithmetic Mean

- The sum of the deviations of the observations from their mean is equal to zero, i.e., $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$

Statement: Given a dataset with n observations $x_1, x_2, \ldots, x_n$, the mean $\bar{x}$ is defined as:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Proof:

- Start with the sum of deviations:

$$\sum_{i=1}^{n}(x_i - \bar{x})$$

- Expand the expression inside the summation:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \bar{x}$$

- Since $\bar{x}$ is a constant:

$$\sum_{i=1}^{n} \bar{x} = n\bar{x}$$

- Substituting and simplifying:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n} x_i - n\bar{x} = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} x_i = 0 \quad \textit{proved}$$

The Sum of Squared Deviations from the Mean

- The Sum of Squared Deviations from the Mean is Minimum.

Statement: We aim to prove that the sum of squared deviations of x_i from their mean $\bar{x}$ is a minimum:

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 \leq \sum_{i=1}^{n}(x_i - a)^2$$

where a is any arbitrary value other than $\bar{x}$.

Proof: The Sum of Squared Deviations from the Mean

- Start with the arbitrary value a:

$$\sum_{i=1}^{n}(x_i - a)^2$$

- Expand the squared terms:

$$\sum_{i=1}^{n}(x_i - a)^2 = \sum_{i=1}^{n} x_i^2 - 2a\sum_{i=1}^{n} x_i + na^2$$

Proof Contn'd...

- The sum of squared deviations from the mean $\bar{x}$:

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

- Subtract the squared deviations from the mean from those of the arbitrary value a:

$$\sum_{i=1}^{n}(x_i - a)^2 - \sum_{i=1}^{n}(x_i - \bar{x})^2 = n(a - \bar{x})^2$$

- Since $n(a - \bar{x})^2 \geq 0$ for all a:

$$\sum_{i=1}^{n}(x_i - a)^2 \geq \sum_{i=1}^{n}(x_i - \bar{x})^2 \quad \textit{Proved}$$

Transformation of Mean under Scaling and Shifting

For a dataset $\{x_1, x_2, \ldots, x_n\}$ with mean $\bar{x}$, if each observation is transformed as:

$$x_i' = k \cdot x_i + c,$$

the new mean becomes:

$$\bar{x}' = k \cdot \bar{x} + c.$$

Proof:

- Start with the original dataset and the new dataset transformed by scaling and shifting.
- Substitute the transformed observations into the mean formula:

$$\bar{x}' = \frac{1}{n} \sum_{i=1}^{n} (k \cdot x_i + c)$$

- Simplify the summation and use the properties of the sum:

$$\bar{x}' = k \cdot \bar{x} + c$$

- **Mean as Projection onto the Span of Ones**. The arithmetic mean can be interpreted as the projection of a data vector onto the span of the vector of ones. (Think!)
- It is sensitive to extreme values (outliers). If there is a very large or very small number in the data set, it can significantly change the mean.
- The mean is best used when the data is symmetrically distributed or when all data points have equal importance and outliers are minimal.

Measures of Central Tendency: Median

Median: Ungrouped Data

Ungrouped Data: Arrange the data in ascending order:

- If n is odd, the median is the middle value.
- If n is even, the median is the average of the two middle values.

Grouped Data: The median is calculated using the formula:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - CF}{f_m} \right) \cdot w$$

where:

- L: Lower boundary of the median class
- N: Total frequency
- CF: Cumulative frequency before the median class
- f_m: Frequency of the median class
- w: Width of the median class

Characteristics of Median:

- It is not affected by outliers or extreme values in the data set.
- The median is especially useful when the data is skewed or contains outliers because it better represents the "typical" value.

Measures of Central Tendency: Mode

Mode: Ungrouped Data

Ungrouped Data: The mode is the most frequently occurring value in the dataset.

Mode: Grouped Data

$$\text{Mode} = L + \left(\frac{f_m - f_{m-1}}{(f_m - f_{m-1}) + (f_m - f_{m+1})} \right) \cdot w$$

where:

- L: Lower boundary of the modal class
- f_m: Frequency of the modal class
- f_{m-1}: Frequency of the class before the modal class
- f_{m+1}: Frequency of the class after the modal class
- w: Width of the modal class

Characteristics of Mode:

- The mode is not affected by outliers since it is based on frequency.
- It is especially useful for categorical data (e.g., color, gender) where you want to know which category is most common.

Finding Mean, Median, and Mode for Grouped Data

Example: Finding Mean, Median, and Mode for Grouped Data

We have the following data on the scores of 40 students in a class:

$$\{50, 55, 58, 60, 63, 65, 68, 70, 72, 73, 75, 78, 80, 82, 85, 87, 88, 90,$$
$$92, 95, 96, 97, 100, 102, 105, 108, 110, 113, 115,$$
$$118, 120, 123, 125, 128, 130, 135, 138, 140, 142, 145\}$$

Step 1: Number of Classes Using Sturges' Rule

Sturges' rule gives the number of classes (k) for grouped data:

$$k = 1 + 3.322 \log_{10} n$$

where n is the number of data points.

For our dataset:

$$n = 40 \quad \Rightarrow \quad k = 1+3.322 \log_{10} 40 = 1+3.322\times1.602 = 1+5.318 = 6.318$$

So, we round it up to $k = 7$ classes.

Example Contn'd...

Step 2: Calculate Class Width

Class width (w) is calculated as:

$$w = \frac{\text{Range}}{k}$$

where the range is the difference between the maximum and minimum values in the dataset.

$$\text{Range} = 145 - 50 = 95$$

$$w = \frac{95}{7} \approx 13.57$$

We round the class width to the nearest whole number, so $w = 14$.

Step 3: Define Class Limits and Boundaries

The first class starts at 50, and the class width is 14. So the class limits and boundaries are:

Class Limits	Class Boundaries
50 – 63	49.5 – 63.5
64 – 77	63.5 – 77.5
78 – 91	77.5 – 91.5
92 – 105	91.5 – 105.5
106 – 119	105.5 – 119.5
120 – 133	119.5 – 133.5
134 – 147	133.5 – 147.5

Step 4: Frequency Distribution Table

Now, let's group the data and create the frequency distribution table. For simplicity, we provide the frequencies for each class:

Class	Frequency(f)	Midpoint(x)	Cumulative Frequency(CF)
$50 - 63$	5	56.5	5
$64 - 77$	7	70.5	12
$78 - 91$	9	84.5	21
$92 - 105$	6	98.5	27
$106 - 119$	5	112.5	32
$120 - 133$	5	126.5	37
$134 - 147$	3	140.5	40

Mean: The mean is calculated as:

$$\bar{x} = \frac{\sum(f \cdot x)}{\sum f}$$

Where f is the frequency and x is the midpoint of the class.

$$\bar{x} = \frac{(5 \cdot 56.5) + (7 \cdot 70.5) + (9 \cdot 84.5) + (6 \cdot 98.5) + (5 \cdot 112.5) + (5 \cdot 126.5) -}{5 + 7 + 9 + 6 + 5 + 5 + 3}$$

$$\bar{x} = \frac{282.5 + 493.5 + 760.5 + 591.0 + 562.5 + 632.5 + 421.5}{40} = \frac{4744}{40} = 118.6$$

$$\text{Mean} = 118.6$$

Median: The median class is the class where the cumulative frequency reaches or exceeds half the total frequency.
The median class is Class 3 (78 - 91), (the first class after CF become greater than $N/2 = 20$). The median is calculated using the formula:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - CF}{f_m} \right) \cdot w$$

where:

- $L = 77.5$ (lower boundary of the median class),
- $N = 40$ (total frequency),
- $CF = 12$ (cumulative frequency before the median class),
- $f_m = 9$ (frequency of the median class),
- $w = 14$ (class width).

Example Contn'd...

$$\text{Median} = 77.5 + \left(\frac{\frac{40}{2} - 12}{9} \right) \cdot 14 = 77.5 + \left(\frac{20 - 12}{9} \right) \cdot 14$$

$$\text{Median} = 77.5 + \left(\frac{8}{9} \right) \cdot 14 = 77.5 + 12.44 = 89.94$$

$$\text{Median} \approx 90$$

Modal Class: The modal class is the class with the highest frequency, which is the class 78 - 91 with frequency $f_1 = 9$.
Formula for Mode:

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \cdot h$$

where:

- L: Lower class boundary of the modal class,
- f_1: Frequency of the modal class,
- f_0: Frequency of the class before the modal class,
- f_2: Frequency of the class after the modal class,
- h: Width of the class.

Example Contn'd...

Given Values:

- $L = 77.5$ (lower class boundary of the modal class),
- $f_1 = 9$ (frequency of the modal class),
- $f_0 = 7$ (frequency of the class before the modal class: 64 - 77),
- $f_2 = 6$ (frequency of the class after the modal class: 92 - 105),
- $h = 14$ (class width).

Substitute the values into the formula:

$$\text{Mode} = 77.5 + \left(\frac{9-7}{2 \times 9 - 7 - 6}\right) \cdot 14$$

Simplifying:

$$\text{Mode} = 77.5 + \left(\frac{2}{18-7}\right) \cdot 14 = 77.5 + \left(\frac{2}{11}\right) \cdot 14$$

$$\text{Mode} = 77.5 + 0.1818 \cdot 14 = 77.5 + 2.545 = 80.045$$

Thus, the **Mode** is approximately:

$\text{Mode} \approx 80$

Example: Dataset Without Outliers

A server processes requests, and the response times (in milliseconds) for a given period are recorded as follows:

$$\text{Response Times: } 120\,\text{ms}, 130\,\text{ms}, 125\,\text{ms}, 140\,\text{ms}, 135\,\text{ms}.$$

Mean:

$$\text{Mean} = \frac{120 + 130 + 125 + 140 + 135}{5} = 130\,\text{ms}.$$

Median:

$$\text{Sorted Data: } 120, 125, 130, 135, 140$$

$$\implies \text{Median} = 130\,\text{ms}.$$

Mode: No repeated value, so no mode.

Example: Dataset With Outliers

Consider a situation where one response time is exceptionally high:

$$\text{Response Times: } 120, 130, 125, 140, 135, 1000.$$

Mean:

$$\text{Mean} = \frac{120 + 130 + 125 + 140 + 135 + 1000}{6} \approx 275\,\text{ms}.$$

Median:

$$\text{Sorted Data: } 120, 125, 130, 135, 140, 1000$$

$$\implies \text{Median} = \frac{130 + 135}{2} = 132.5\,\text{ms}.$$

Mode: No repeated value, so no mode. **Interpretation**: The median remains relatively stable, as it is less sensitive to extreme values.

Other Types of Measures of Central Tendency

1. Geometric Mean

Definition: The *n*th root of the product of *n* values.
Formula:

$$\text{Geometric Mean} = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}}$$

Example: For the dataset $[1, 3, 9, 27]$:

$$\text{Geometric Mean} = (1 \cdot 3 \cdot 9 \cdot 27)^{\frac{1}{4}} = \sqrt[4]{729} = 9$$

Characteristics:

- Useful for data with exponential growth (e.g., populations, compound interest).
- Less sensitive to extreme values than the arithmetic mean.
- Suitable for multiplicative relationships (e.g., rates of change).

Definition: The reciprocal of the arithmetic mean of reciprocals.

Formula:

$$\text{Harmonic Mean} = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

Example: For the dataset $[2, 3, 4]$:

$$\text{Harmonic Mean} = \frac{3}{\frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{3}{\frac{13}{12}} = \frac{36}{13} \approx 2.77$$

Characteristics:

- Best for data representing rates, such as speed or efficiency.
- Highly influenced by small values.

3. Weighted Mean

Definition: Averages data points with different weights.
Formula:

$$\text{Weighted Mean} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

Example: For the dataset $[10, 20, 30]$ with weights $[1, 2, 3]$:

$$\text{Weighted Mean} = \frac{(10 \cdot 1) + (20 \cdot 2) + (30 \cdot 3)}{1 + 2 + 3} = \frac{140}{6} \approx 23.33$$

Characteristics:

- Useful when data points have different levels of importance or represent different sample sizes.
- Common in grading systems and survey data.

4. Trimmed Mean

Definition: Removes extreme values before calculating the mean.
Example: For the dataset $[1, 2, 3, 100]$, trim 25%:

$$\text{Trimmed Mean} = \frac{2+3}{2} = 2.5$$

Characteristics:

- Reduces the influence of outliers and extreme values.
- Useful in situations where data may contain errors or extreme values.

5. Quadratic Mean (Root Mean Square)

Definition: Square root of the mean of squared values.
Formula:

$$\text{Quadratic Mean} = \sqrt{\frac{\sum_{i=1}^{n} x_i^2}{n}}$$

Example: For the dataset $[3, 4, 5]$:

$$\text{Quadratic Mean} = \sqrt{\frac{3^2 + 4^2 + 5^2}{3}} = \sqrt{\frac{50}{3}} \approx 4.08$$

Characteristics:

- Useful when data represents squared values, such as deviations or in signal processing.
- Larger values can have more influence than smaller values.

6. Power Mean

Definition: A generalization of the mean.

Formula:

$$\text{Power Mean} = \left(\frac{\sum_{i=1}^{n} x_i^p}{n} \right)^{\frac{1}{p}}$$

Properties:

- For different values of p, the power mean interpolates between various means:
  - $p = 1$: The power mean becomes the **arithmetic mean**.
  - $p = 0$: The power mean approaches the **geometric mean**.
  - $p = -1$: The power mean becomes the **harmonic mean**.
  - $p = 2$: The power mean becomes the **quadratic mean (root mean square)**.
- When $p \to \infty$: The power mean approaches the **maximum value** of the dataset.
- When $p \to -\infty$: The power mean approaches the **minimum value** of the dataset.

Characteristics:

- Flexible mean that generalizes other means depending on the value of p.
- Useful for situations where different types of averaging are needed.

7. Midrange

Definition: The average of the minimum and maximum values in a dataset.

Formula:

$$\text{Midrange} = \frac{\text{Maximum Value} + \text{Minimum Value}}{2}$$

Characteristics:

- Not robust to outliers, as it only considers the minimum and maximum values.
- May not represent the data well if there are outliers.
- Useful for a quick estimate of central tendency.

Exploratory Data Analysis (EDA)

Definition: EDA involves preliminary checks to understand the data's nature, identify anomalies, and select appropriate statistical methods.

Key Tools:

- **Stem-and-Leaf Plot:** Visualizes data distribution in a compact form.
- **Box Plot:** Highlights data spread, median, and potential outliers.

Historical Note: EDA was introduced by John Tukey in 1977.